

Sprint 2 Report

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Sprint Objectives

The goal of Sprint 2 was to:

- Continue development of the robot control software
- Modify the drive logic to support the updated front-wheel tank drive configuration
- Demonstrate obstacle detection and avoidance behaviour using grayscale camera vision processing
- Finalize the mechanical redesign of the chassis, ball mount, and camera mount
- Prepare the mechanical components for future manufacturing

By the end of the sprint, we aimed to demonstrate a working prototype capable of basic obstacle detection and avoidance while completing the mechanical design components required for the redesigned smart car platform.

Steps to Achieve the Sprint Goal

1. Software Development & Drive Logic Updates

Software development continued to expand the functionality implemented during Sprint 1.

- Updated the motor control logic to reflect the new mechanical configuration where the rear motors are removed.
- Modified the codebase so that the robot operates using only the two front motors for movement and turning.
- Verified that the updated drive configuration correctly performs tank-style turning using the front wheels.

During testing, the robot successfully performs the following behaviours:

- Moves forward under normal conditions
- Detects obstacles using grayscale camera image processing
- Stops when an obstacle is detected within the camera field of view
- Performs a turning manoeuvre using the two front wheels to redirect the vehicle away from the obstacle

These behaviours were validated on the physical robot and recorded as part of the sprint demonstration. This confirms that the updated drive configuration and vision processing logic operate correctly with the modified mechanical design.

2. Mechanical Redesign Completion

Mechanical redesign work continued from Sprint 1 and was finalized during this sprint. The following components were completed using SolidWorks:

Chassis Design

- Completed the full SolidWorks model of the redesigned chassis.
- The design reuses existing mounting holes from the original platform to simplify assembly.
- The chassis removes support for the rear wheels and integrates mounting features for the rear ball mount.

Rear Ball Mount

- Finalized the SolidWorks model for the rear ball mount assembly.
- The mount provides a stable support point to replace the removed rear wheels.
- Mounting holes were designed to integrate directly with the redesigned chassis.

Camera Mount

- Completed the design of the fixed camera mount.
- The design replaces the original three-servo camera assembly.
- The mount secures the camera in a fixed forward-facing orientation suitable for future AprilTag detection.

All mechanical components were exported as STL files and prepared for 3D printing. Manufacturing has not yet begun, but the designs are ready for fabrication in the next stage of the project.

Demo Link

A demonstration video is provided below showing the robot performing basic motion and obstacle detection using the updated drive configuration.

The video demonstrates forward motion using the two front motors, obstacle detection using grayscale camera processing, a stop response when an obstacle is detected in the camera view, and turning behaviour using the front-wheel tank drive configuration.

Link:

<https://uwoca.sharepoint.com/:v:/s/MechaTron/IQA9llykdzBbRKuAWfQGD0KoAckAaB3q-SIHHPgtMaTZC7c?e=NtDYGd>

Review & Retrospective

1. What Went Well During This Sprint?

Drive System Adaptation

The transition from the original four-wheel configuration to the two-front-wheel drive system was successfully implemented. Updating the motor control code allowed the robot to move and turn reliably using only the front motors. This confirms that the modified drivetrain concept is viable for the redesigned platform.

Obstacle Detection Behaviour

The robot successfully demonstrated obstacle detection and response behaviour using grayscale camera processing. The system detects obstacles within the camera's field of view, stops the vehicle, and performs a turn to redirect the robot. This represents an improvement over the previous sprint, where only forward motion and stopping behaviour had been implemented.

Mechanical Design Completion

All major mechanical redesign components were completed during this sprint. The chassis, ball mount, and camera mount were finalized in SolidWorks and exported as STL files for printing. Completing the CAD stage allows the team to move forward with fabrication and physical integration.

2. What Didn't Go Well / What Can Be Improved?

Limited Vision System Integration

Although progress was made in developing the AprilTag detection code, the vision system has not yet been integrated with the robot's navigation logic. As a result, the robot currently relies only on basic vision-based obstacle detection and turning behaviour.

Causes: Time during the sprint was primarily focused on adapting the drivetrain code and completing the mechanical design.

Improvements: Next sprint will focus on integrating AprilTag detection into the control loop and enabling the robot to respond to visual markers.

Manufacturing Not Yet Completed

Although the mechanical designs were completed, the components have not yet been physically manufactured.

Improvements: Begin 3D printing the chassis, ball mount, and camera mount early in the next sprint to allow sufficient time for testing and iteration.

Testing Environment Limitations

Testing was conducted in a limited indoor environment, which restricted the ability to perform extended navigation trials.

Improvements: Set up a more structured testing environment for AprilTag detection and navigation testing during the next sprint.

Plan for the Next Sprint

- Begin 3D printing the chassis, ball mount, and camera mount components.
- Assemble the redesigned mechanical structure on the robot.
- Continue development of the AprilTag detection system using the grayscale camera feed.
- Integrate AprilTag detection with the robot navigation logic.
- Expand testing to include vision-based navigation behaviour.